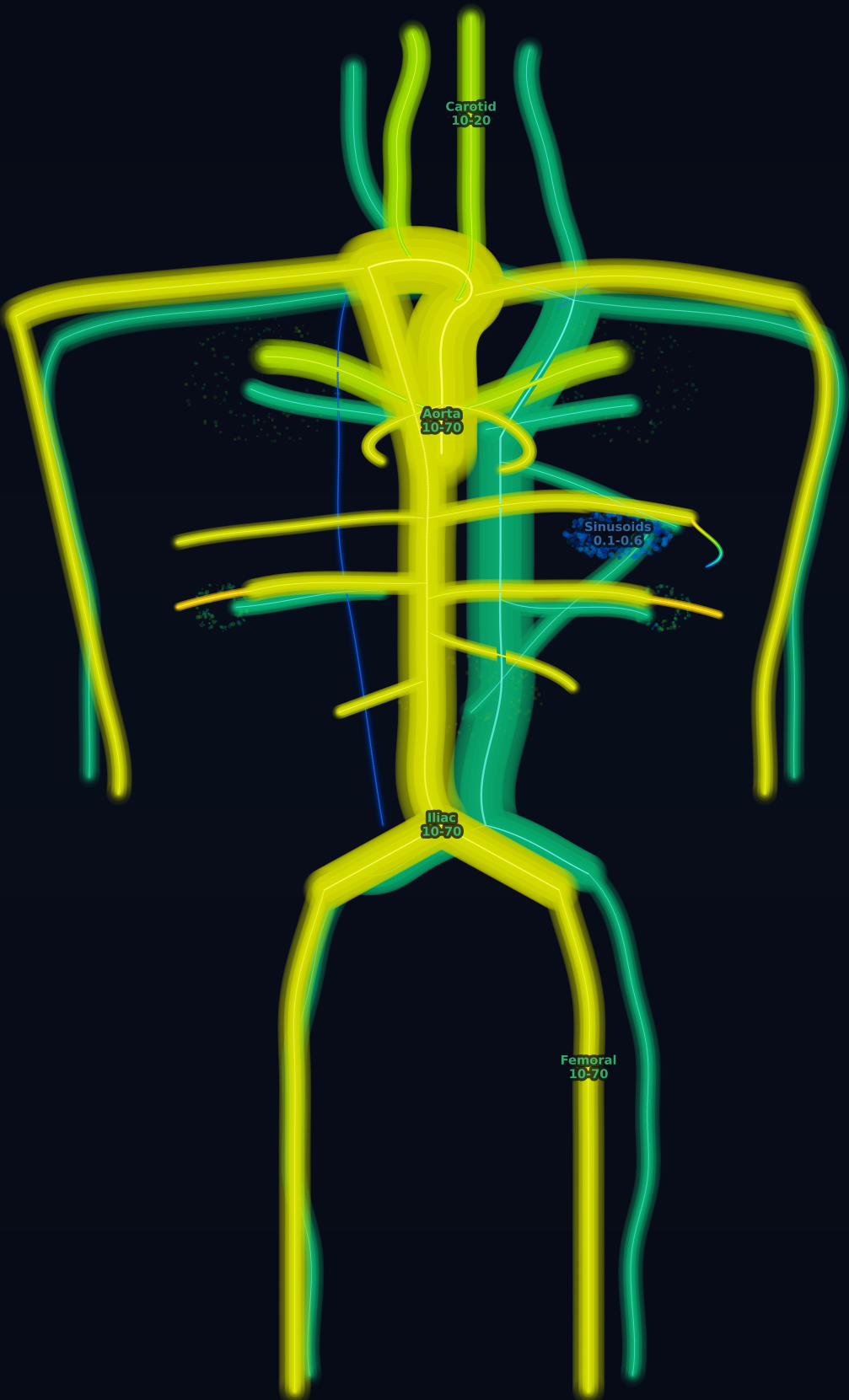


Wall Shear Stress Comparison — Hepatocellular Carcinoma

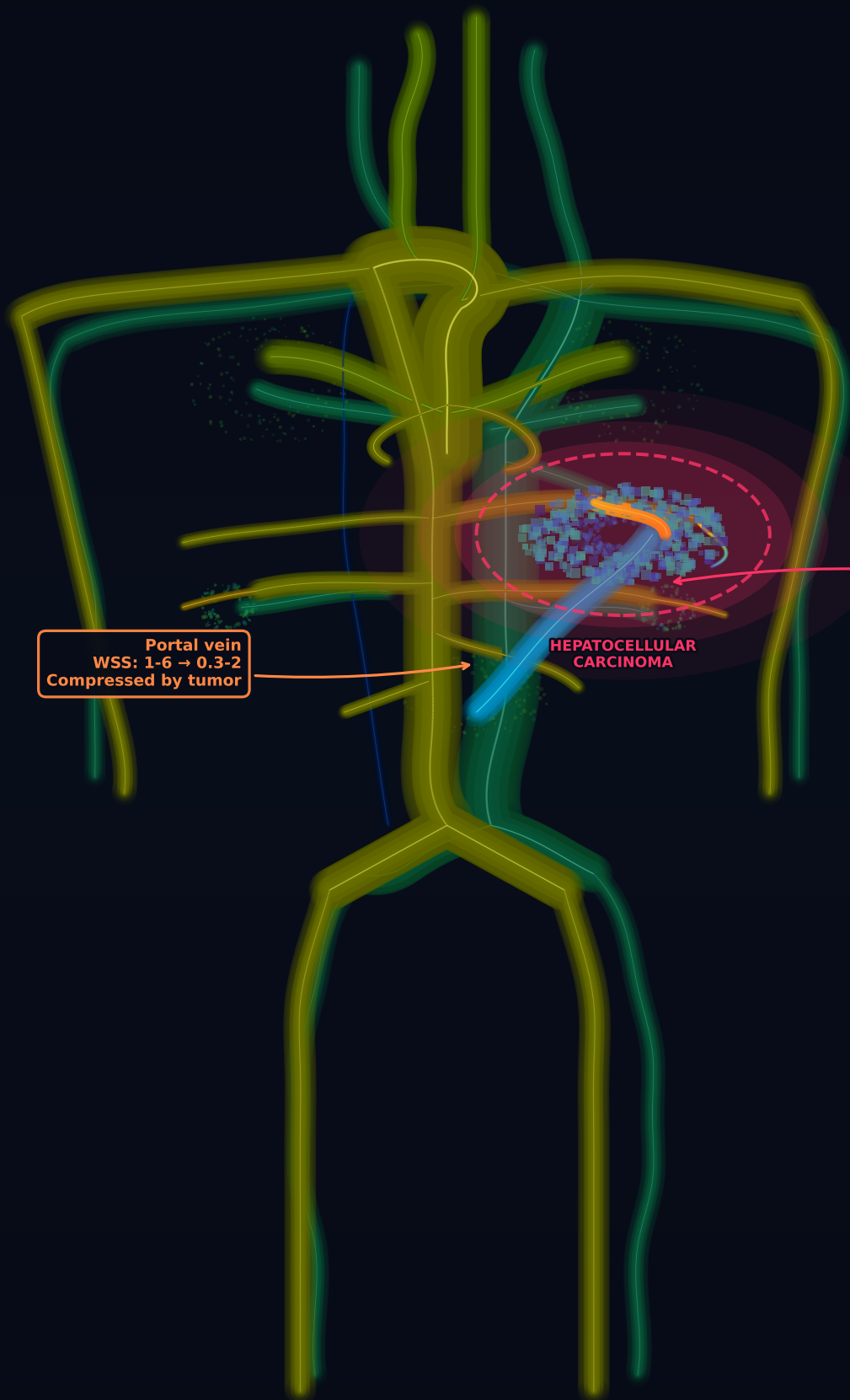
HEALTHY

Normal physiological WSS



HEPATOCELLULAR CARCINOMA

HCC arterIALIZATION disrupts hepatic sinusoidal flow



WSS Comparison
Healthy vs Hepatocellular Carcinoma

Venous

Iliac

Femoral

Renal Art.

Hepatic Sin.

Pulmonary

Coronary

Aorta

Carotid

WSS (dyne/cm²)

Healthy
Pathological (↑)
Pathological (↓)



Wall Shear Stress (dyne/cm²)

Arterial (10-70) Venous (1-6) Lymphatic (0.1-0.6) Stenosis
Plaque site Tumor

Data: Modh et al., "Resolving the Biomechanical Blind Spot in Nanomedicine Translation" — ACS Nano (2025)